

Summary

Data scientist focused on applied machine learning and predictive modeling. My work has centered on model selection, tuning, evaluation, and building scalable data workflows.

Projects

Aspect-Based Sentiment Analysis of Google Reviews (2026)

(Capstone Project, [Paper](#), [GitHub](#))

Businesses receive thousands of reviews but lack tools to extract actionable, aspect-level feedback at scale. This project built and benchmarked an end-to-end NLP pipeline to solve that.

- ▶ Processed 70M+ raw reviews (25GB+) and engineered a weak supervision pipeline to generate 14 aspect-sentiment labels at scale
- ▶ Benchmarked TF-IDF, fine-tuned RoBERTa, and a zero-shot LLM for aspect-level sentiment classification
- ▶ Zero-shot LLM outperformed all trained models on human-labeled ground truth (F1 0.757 vs. 0.605), exposing weak supervision's performance limits

Amazon Sports Products Recommender System (2025)

(Graduate Project, [GitHub](#))

E-commerce platforms lose sales when they can't surface the right products to the right users. This project built a recommendation system across 4M+ Amazon reviews to deliver personalized product rankings.

- ▶ Mitigated 99.998% data sparsity by engineering user and item behavioral features, then benchmarking five approaches from global baseline to SVD matrix factorization
- ▶ Combined collaborative filtering (SVD) with Ridge regression to balance personalization and cold-start coverage
- ▶ Hybrid model (0.6 SVD + 0.4 Ridge) reduced RMSE by 31.5% over baseline, outperforming benchmarked approaches

NYC Yellow Taxi Fare Prediction (2025)

(Graduate Project, [My Contributions](#) | [GitHub](#))

Accurate fare estimation is critical for ride-hailing platforms to maintain rider trust and operational efficiency. This project built and validated a fare prediction model across 150M+ NYC taxi records, earning 3rd place in a class competition.

- ▶ Engineered trip duration, distance, and toll features using Spark, identifying trip duration as a key fare driver despite low correlation
- ▶ Trained a distributed LightGBM regressor, reducing RMSE by 49% over baseline, achieving R² 0.94 with near-zero train/test gap confirming strong generalization

Education

M.S. in Data Science	2026
<i>University of California, San Diego</i>	La Jolla, CA
GPA: 3.81	
B.S. in Management Information Systems, magna cum laude	2024
<i>California State University, Sacramento</i>	Sacramento, CA
A.S. in Accounting A.S. in Business Administration	2022

Experience

DBI Beverage	2014 - 2020
<i>Operations Team Lead</i>	Stockton, CA
▶ Performed data analysis using WMS, inventory, shipment, and spreadsheet data to investigate discrepancies, conduct root cause analysis, and support inventory decisions ▶ Led warehouse staff training during implementation of a new Warehouse Management System, supporting the transition from RF scanners to iPad-based workflows ▶ Improved operations across receiving, storage, FIFO, and replenishment, contributing to a 12% improvement in inventory accuracy	

Skills

Programming: Python, SQL

Machine Learning: Scikit-learn, PyTorch, XGBoost, LightGBM, Hugging Face Transformers

Methods: Statistics, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning

Data & Tools: Pandas, NumPy, Apache Spark, PostgreSQL, Git, Jupyter